(Refer Glasserman-Sec 2.2, Seydel(4e)-Sec 2.2)

We assume the availability of a sequence $U_1, U_2, \cdots$, of independent and identically distributed uniform variates (which you can generate using the methods known to you, but you have to keep in mind that we need 'good' uniform variates).

1. Consider the probability density function

$$f(x) = \begin{cases} 3(1-x)^2 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

   (a) Generate $X_1, X_2, \ldots, X_N$ from the above distribution for $N = 10, 100, 1000, 10000, 100000$.

   (b) For each value of N, plot the empirical CDF (definition is given below) of these generated values, and the actual distribution function (using the above formula).
   Defn: Let $x_1, x_2, \ldots, x_N$ be sample observations. Then the empirical CDF at $x \in \mathbb{R}$ is defined by

$$F_N(x) = \frac{\#\{i : x_i \leq x\}}{N}.$$

   (c) Provide the corresponding values of the sample mean and variance. Compare the values of sample mean and variance with their population counterpart to see if the desired convergences happen.

2. Consider the cumulative distribution function:

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ 1 - e^{-x} & \text{if } 0 < x \leq 1 \\ 1 - e^{-(2x-1)} & \text{if } x > 1. \end{cases}$$

   Do an exercise similar to the one given in Item-(1)[a-c] above.

3. Using the algorithm to generate random variables from a discrete distribution, generate 100000 random numbers from a discrete uniform distribution on $\{1, 3, 5, \ldots, 9999\}$. Tabulate the frequency of each observed values.

4. Now consider the problem of sampling X (whose distribution is F) conditional on $a < X \leq b$ with $F(a) < F(b)$. Generate a random sample from $\mathcal{U}[0.3, 0.7]$, by conducting an exercise similar to the one given in Item-(1)[a-c] above.

Also, observe the variates for any pattern, repetition, etc. Put all your observations in the report.